

UCS420: COGNITIVE COMPUTING

L	T	P	Cr
2	0	2	3.0

Course Objectives: This course will provide advanced students in cognitive science and computer science with the skills to develop computational models of human cognition, giving insight into how people solve challenging computational problems, as well as how to bring computers closer to human performance.

Introduction to Computational Modeling: Feedforward neural networks, Hopfield networks and Boltzmann machines, Simple Recurrent Networks (Elman), Growth models (Cascade correlation), Spiking Neural Networks, Edged AI.

Model Optimization: Techniques like pruning, quantization, and knowledge extraction in AI models, lowering energy consumption.

Reasoning and Learning: Introduction to Bayesian Networks, Semantics of Bayesian Networks, Case based reasoning, Analogical reasoning, Constraint reasoning and Meta reasoning, Concept learning, Explanation based learning.

Computational Linguistics: Syntax and Parsing, Semantic Representation, Semantic Interpretation, Language Generation, Knowledge Extraction and Summarization, Sentiment Analysis, Interactive Fiction.

Advance analytics: Big data and Cognitive computing, Cognitive IoT, Predictive Analytics, Text Analytics, Image Analytics, Speech Analytics.

Case-Study: Introduction to IBM Watson, Components of Deep QA: Building dataset, Question Analysis, Hypothesis generation, scoring and estimation in Watson services.

Laboratory Work:

- Optimizing a spiking neural network in NengoDL
- Legendre Memory Units in NengoDL
- Optimizing a cognitive model in NengoDL
- Optimizing a cognitive model with temporal dynamics in NengoDL
- Build a Text summarizer using Python toolbox

Course Learning Outcomes (CLOs) / Course Objectives (COs):

After the completion of the course, the student will be able to:

1. Deploy language agents for language processing workflows.
2. Design Image information extraction frameworks for different applications.
3. Develop spiking neural network models for various cognitive tasks.
4. Implement various analytics tools in related areas of computer science.

Text Books:

1. Judith S., Kaufman M., and Bowles A., Cognitive Computing and Big Data Analytics, Wiley,(2005), 1st ed.

2. Finger P., Cognitive Computing: A Brief guide for Game Changers, , Meghan-Kiffer Press,(2014),1st ed.

Reference Books:

1. Miller J., Learning IBM Watson Analytics, Packt Publishers,(2016),1st ed.
2. Hashmi A. and Masood A., Cognitive Computing Recipes: Artificial Intelligence solutions using Microsoft Cognitive Services and Tensorflow, Apress, (2019), 1st ed.
3. Kashyap P., Machine Learning for Decision Makers: Cognitive Computing Fundamentals for better decision making,,Apress,(2018), 1st ed.